

D887

Molecular Biology

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

In a teacher-prep lab, candidates review a classic experiment in which bacteria were grown on heavy nitrogen and then moved to light nitrogen for one round of replication. The experiment was published decades ago and is often shown with a colorful animation. After one round, all DNA is of intermediate density. What model does this support?

- A. Replication only on the leading strand
- B. Dispersive replication, because density is always intermediate in every round
- C. Conservative replication
- D. Semiconservative replication

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2.

A student analyzing DNA from a sample collected on a school trip to a coastal marsh finds that 30% of the bases are adenine. The sample was stored in a freezer for two weeks. What percentage of the bases is guanine?

- A. 70%
- B. 30%
- C. 40%
- D. 20%

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3. [TRANSFER — reversed stem / different variable]

A double-stranded DNA sample is 18% cytosine. What percentage is adenine?

- A. 18%
- B. 82%
- C. 64%
- D. 32%

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4.

A teacher writes a DNA template strand on the board: 3'-TACGGA-5'. She uses a new set of colored markers and students copy it into notebooks. What mRNA is transcribed?

- A. 5'-AUGCCU-3'
- B. 5'-ATGCCT-3'
- C. 3'-UACGGA-5'
- D. 5'-TACGGA-3'

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5. [TRANSFER — reversed stem / different variable]

An mRNA reads 5'-AUGGCA-3'. What is the DNA template strand?

- A. 3'-TACCGT-5'
- B. 3'-UACCGU-5'
- C. 3'-AUGGCA-5'
- D. 5'-ATGGCA-3'

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6.

A lab on sickle cell uses a gene whose start is 5'-AUG GAA CUU...-3' in the mRNA. The lab was adapted from a university outreach kit. A single extra base is inserted right after AUG. What is the most likely effect?

- A. Only one amino acid changes.
- B. No effect, because one base is too small to matter.
- C. A frameshift that changes every codon after the insertion, often making a nonfunctional protein.
- D. The protein becomes longer by exactly one amino acid.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7. [TRANSFER — reversed stem / different variable]

A substitution changes the third base of a codon from GAA to GAG. Both code for glutamic acid. What type of mutation is this?

- A. A nonsense mutation
- B. A frameshift mutation
- C. A silent mutation
- D. A missense mutation

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8.

E. coli are grown in a flask that has lactose and no glucose. The flask sits on a shaker that was repaired last month. In what state is the lac operon?

- A. Highly expressed
- B. Expressed at a low level only, because lactose alone is not enough.
- C. Off, because the repressor binds lactose and blocks transcription.
- D. Off, because the operon only works when glucose is present.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9.

Students study a nerve cell model on the district's new simulation platform. The sodium-potassium pump moves 3 Na⁺ out and 2 K⁺ in, both against their gradients. What does the pump require?

- A. Nothing, because it is a membrane protein, so it is passive.
- B. ATP, because this is active transport
- C. Nothing, because ions always diffuse into cells.
- D. Energy only for the sodium, not the potassium

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10.

A teacher candidate builds a lesson on cellular respiration using a video filmed at a cycling race. The video lasts eight minutes. Which stage produces most of the ATP from one glucose molecule?

- A. Glycolysis
- B. The citric acid cycle
- C. Oxidative phosphorylation
- D. Fermentation

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11. [TRANSFER — reversed stem / different variable]

A cell suddenly runs out of oxygen. Which process stops first?

- A. Glycolysis
- B. Electron transport, because oxygen is the final electron acceptor
- C. Protein synthesis
- D. All ATP production immediately

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12.

A student asks why a muscle cell and a nerve cell from the same person look so different. The student is in the honors program and wants to be a doctor. What is the best explanation?

- A.** The cells differ only because of what the person eats.
- B.** Both cells have the same genes, but they express different sets of them.
- C.** Nerve cells lose the genes they do not need.
- D.** Each cell type has a different set of genes.

Target word _____	The one fact that decides _____	Answer ____	☐ sure ☐ unsure ☐ guess
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STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer D Concept 1

In a teacher-prep lab, candidates review a classic experiment in which bacteria were grown on heavy nitrogen and then moved to light nitrogen for one round of replication. ~~The experiment was published decades ago and is often shown with a colorful animation.~~ After one round, all DNA is of intermediate density. What model does this support?

Target word: model of replication **Decides:** Hybrid density after one round means each molecule has one old strand.

A	Confusion set	Strand synthesis mode does not explain density bands.
B	Absolute language	Dispersive is ruled out after the second round shows a light band.
C	Inverted operation	Conservative would give one heavy and one light band.
D	Correct	Each daughter molecule keeps one parent strand and gains one new one.

2. Answer D Concept 2

A student analyzing DNA from a sample ~~collected on a school trip to a coastal marsh~~ finds that 30% of the bases are adenine. ~~The sample was stored in a freezer for two weeks.~~ What percentage of the bases is guanine?

Target word: percent G **Decides:** $A = T = 30\%$, so $G + C = 40\%$.

A	Inverted operation	That is everything except A.
B	Confusion set	Sets G equal to A instead of to C.
C	Dropped element	That is G + C together.
D	Correct	$100 - 30 - 30 = 40$; $G = C = 20\%$.

3. Answer D Concept 2 · transfer

A double-stranded DNA sample is 18% cytosine. What percentage is adenine?

Target word: percent A **Decides:** $C = G = 18\%$.

A	Confusion set	A pairs with T, not C.
B	Inverted operation	Subtracts only C from 100.
C	Dropped element	That is A + T together.
D	Correct	$100 - 18 - 18 = 64$; $A = T = 32\%$.

4. Answer A Concept 3

A teacher writes a DNA template strand on the board: 3'-TACGGA-5'. ~~She uses a new set of colored markers and students copy it into notebooks.~~ What mRNA is transcribed?

Target word: mRNA sequence **Decides:** mRNA is complementary to the template, with U for T.

A	Correct	Pair T-A, A-U, C-G, G-C and write 5' to 3'.
B	Dropped element	Complementary, but keeps T instead of U.
C	Confusion set	Replaces T with U in the template instead of pairing.
D	Inverted operation	Copies the template instead of complementing it.

5. Answer A Concept 3 • transfer

An mRNA reads 5'-AUGGCA-3'. What is the DNA template strand?

Target word: solve for template **Decides:** Template is complementary to mRNA and antiparallel.

A	Correct	Pair A-T, U-A, G-C, C-G.
B	Confusion set	Uses U, which is not in DNA.
C	Inverted operation	Copies the mRNA instead of pairing.
D	Adjacent category	That is the coding strand, not the template.

6. Answer C Concept 4

A lab on sickle cell uses a gene whose start is 5'-AUG GAA CUU...-3' in the mRNA. ~~The lab was adapted from a university outreach kit.~~ A single extra base is inserted right after AUG. What is the most likely effect?

Target word: effect of insertion **Decides:** An insertion shifts every codon downstream.

A	Confusion set	That describes a substitution.
B	Absolute language	One inserted base changes the reading frame.
C	Correct	Codons are read in threes, so an insertion shifts the frame.
D	Adjacent category	One base is not a full codon; the frame shifts.

7. Answer C Concept 4 • transfer

A substitution changes the third base of a codon from GAA to GAG. Both code for glutamic acid. What type of mutation is this?

Target word: classify the mutation **Decides:** Same amino acid.

A	Adjacent category	A nonsense mutation creates a stop codon.
B	Confusion set	Substitutions do not shift the frame.
C	Correct	The codon changes but the amino acid does not.
D	Inverted operation	Missense changes the amino acid, which does not happen here.

8. Answer A Concept 5

E. coli are grown in a flask that has lactose and no glucose. ~~The flask sits on a shaker that was repaired last month.~~ In what state is the lac operon?

Target word: lac operon state **Decides:** Lactose removes the repressor; low glucose activates CAP.

A	Correct	No repressor on the operator and CAP-cAMP bound.
B	Dropped element	Ignores that glucose is also absent.
C	Inverted operation	Lactose (as allolactose) pulls the repressor off the DNA.
D	Confusion set	High glucose lowers expression.

9. Answer B Concept 6

Students study a nerve cell model ~~on the district's new simulation platform~~. The sodium-potassium pump moves 3 Na⁺ out and 2 K⁺ in, both against their gradients. What does the pump require?

Target word: energy requirement **Decides:** Movement against gradients.

A	Confusion set	Channels and carriers can be passive or active; direction decides.
B	Correct	Moving ions against gradients needs energy.
C	Absolute language	Ions move along their gradients, which can go in or out.
D	Dropped element	Both ions move against their gradients using one ATP cycle.

10. Answer C Concept 7

A teacher candidate builds a lesson on cellular respiration ~~using a video filmed at a cycling race. The video lasts eight minutes~~. Which stage produces most of the ATP from one glucose molecule?

Target word: largest ATP source **Decides:** Oxidative phosphorylation produces the most ATP.

A	Inverted operation	Glycolysis nets only 2 ATP.
B	Confusion set	It makes 2 ATP (or GTP) per glucose; its main output is NADH and FADH ₂ .
C	Correct	The electron transport chain and ATP synthase make most of the ATP.
D	Adjacent category	Fermentation only regenerates NAD ⁺ ; it makes no extra ATP.

11. Answer B Concept 7 · transfer

A cell suddenly runs out of oxygen. Which process stops first?

Target word: what requires O₂ **Decides:** Oxygen is the final electron acceptor.

A	Inverted operation	Glycolysis does not need oxygen.
B	Correct	With no oxygen, electrons back up in the chain.
C	Adjacent category	Not the step that uses oxygen directly.
D	Absolute language	Glycolysis with fermentation still makes some ATP.

12. Answer B Concept 8

A student asks why a muscle cell and a nerve cell from the same person look so different. ~~The student is in the honors program and wants to be a doctor.~~ What is the best explanation?

Target word: cause of cell differences **Decides:** Same genome, different genes expressed.

A	Adjacent category	Diet can influence some gene expression, but it does not explain why cell types differ.
B	Correct	Differentiation is differential gene expression.
C	Inverted operation	Genes are silenced, not removed.
D	Confusion set	Nearly all body cells share the same genome.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	5	
Wrong stem number	A figure copied from the prompt that is not the answer.	0	
Adjacent category	The right idea in the wrong bucket.	6	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	10	
Absolute language	Always, never, guarantee — where judgment applies.	4	
Confusion set	Every option is real; the test is exact role-matching.	11	
Camouflage	Answers a stem fact that was never the question.	0	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Semiconservative replication and strand direction	Each new DNA molecule keeps one old strand; polymerase adds nucleotides only to the 3' end, so the leading strand is continuous and the lagging strand is made in Okazaki fragments.	Thinking both strands are copied continuously, or that replication makes one all-new molecule.	Okazaki fragments, 5' to 3' direction, or the Meselson-Stahl result.
2	Chargaff's base-pairing rule	In double-stranded DNA, A = T and G = C, so the percentages of A + G equal 50%.	Setting A equal to G, or making all four bases equal.	One base percentage given and another asked.
3	Transcription uses the template strand	RNA polymerase reads the template strand 3' to 5' and builds mRNA 5' to 3' that matches the coding strand, with U in place of T.	Copying the template strand directly or keeping T in the mRNA.	A DNA strand labeled template vs. coding, and an mRNA to write.
4	Point mutations vs. frameshifts	A single-base insertion or deletion shifts the reading frame and changes every codon after it; a substitution changes at most one codon and may be silent.	Thinking all single-base changes have equally small effects.	Insertion or deletion vs. substitution, and where in the gene it occurs.
5	The lac operon	The lac operon is expressed most when lactose is present (repressor removed) and glucose is low (CAP-cAMP activates).	Thinking lactose alone turns the operon fully on, or that the repressor binds lactose to block transcription.	Levels of glucose and lactose in the medium.
6	Active vs. passive transport	Moving a substance against its concentration gradient needs energy, such as ATP in the sodium-potassium pump (3 Na ⁺ out, 2 K ⁺ in); passive transport moves down the gradient.	Calling any protein-assisted transport active.	Direction relative to the gradient and whether ATP is used.
7	Cellular respiration yields	Glycolysis in the cytoplasm nets 2 ATP; most ATP comes from oxidative phosphorylation in the mitochondria, which needs oxygen as the final electron acceptor.	Thinking glycolysis or the citric acid cycle makes most of the ATP, or that oxygen is used in glycolysis.	Where ATP is made, or what happens when oxygen runs out.
8	Differentiation is differential gene expression	Specialized cells in one body have the same genome; they differ because they express different genes.	Thinking cells lose or gain genes as they specialize.	Two cell types from the same organism compared.

Course scope (public description): Molecular Biology is designed to deepen the understanding of the intricate processes that govern life at the molecular level. This course explores the structure and function of key molecules involved in DNA replication, transcription, translation, and gene regulation. It delves into the mechanisms of cellular transport, differentiation, and regulation, examining how cells maintain their internal environment and respond to external signals. The biochemical pathways essential for metabolism and energy flow, including glycolysis, the citric acid cycle, and oxidative phosphorylation, are thoroughly covered. This course equips

educators with the knowledge and skills to effectively teach molecular biology concepts in the classroom, emphasizing evidence-based applications and critical thinking.